

Improving Occupant Packaging Posture Prediction Through Integration of Simulation and Real Data

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Abstract. This paper presents a process for enhancing driver posture prediction in digital human modelling (DHM) tools by integrating real-world data from driver posture studies. The process leverages key data points such as seat position, steering wheel position, and eye point coordinates to position manikins and extract joint angles for ergonomic analysis. A test study involving 49 drivers was conducted, revealing significant variations in joint angles across different statures. These findings were used to develop stature-based strategies that demonstrated improved predictive accuracy for short and tall stature groups compared to the existing strategy in the DHM tool IPS IMMA. While the results highlight the potential benefits of this approach, limitations such as refined manikin body meshes, seat property considerations, and broader vehicle model validation are recommended. Overall, this method offers a promising solution for addressing incomplete datasets in occupant packaging studies, contributing to the development of more ergonomic and safer vehicle designs.

Keywords: Occupant packaging, digital human modelling, simulation, dataset, process

1 Introduction

Different driver posture models have been developed through data-driven and optimization approaches and implemented in digital human modelling (DHM) tools [1], [2]. Data-driven models predict human body joint angles or positions based on various human and vehicle parameters, while optimization methods aim to minimize discomfort by decreasing deviations from expected neutral or optimal joint angles. Evaluating and validating these models is essential in occupant packaging to be able to ensure ergonomic requirements such as comfort, operating controls, and visibility for a diverse population [3]. However, obtaining comprehensive datasets for occupant packaging validation studies can be challenging in many research and design scenarios. This often results in insufficient and limited data [4], which hinders thorough analysis and enhancement of prediction models in DHM tools. An issue can arise when the prescribed preferred driver angles in the simulation tool are not derived from a similar biomechanical model. Joint chains such as the spine might be simplified in simula-

tions, or the joint angle data can be given only in a two-dimensional sagittal plane. In addition, the validity of laboratory studies for predicting driving posture in real vehicles can, in some cases, be uncertain, and the possibility of doing detailed driving posture studies in real vehicles is limited. However, data of coordinates for the seat position, steering wheel position and driver eye point are relatively easy to measure, even in real vehicles. Still, such data points are not measured directly on the driver, except for the eye-point, which introduces the additional challenge of accurately estimating the mid-hip to h-point offset. This offset measurement, or prediction of it, is often used as input for future posture prediction studies where inverse kinematics posture the digital human models, so-called manikins, towards a preferred driving posture [5].

This paper introduces a process where key data from driver studies in real vehicles, such as coordinates for the seat position, steering wheel position, and driver eye point, is used to position manikins and extract their joint angles. These extracted joint angles can be used to analyze posture and position differences in the population and consequently generate improved posture predictions in less constrained simulation settings for future vehicle ergonomics design. While similar processes have been used in the past [6], the most significant difference lies in its implementation in a DHM tool, allowing the consideration of additional aspects such as anthropometric body meshes or collision avoidance with vehicle geometries.

2 Method

The core idea of this process is to leverage existing real-world data as input for simulations alongside simulations data to fill in missing posture and position data, enriching the dataset and enabling a more comprehensive analysis of occupant packaging scenarios. The process contains three main steps (Figure 1). The first is collecting data on test persons, such as anthropometry and preferred positions of seat (H-point), seat back angle, steering wheel, and eye point. Next, manikins based on each test person's anthropometry are generated. Thirdly, each manikin is positioned in the preferred position of the test person it is representing together with general constraints such as feet on pedals and no collision with surrounding geometry, i.e. roof, dashboard and console. The positioning of the hip and buttock area as well as the back of each manikin are done using a body shape alignment approach [6]. Each manikin's buttock and back surface are aligned with an H-point template positioned in the preferred H-point position of the test person it is representing. From each positioned manikin, body joint angles, positions, and distance to specific vehicle geometries are extracted.

Based on the complete data set containing both anthropometric data and their corresponding preferred body joint angles, statistical analysis can be done to define vehicle or even population-specific optimal joint angles. Lastly, the new optimal joint angles can be used in less constrained simulation settings for future vehicles where the exact position of the seat, steering wheel, and eye point is unknown.

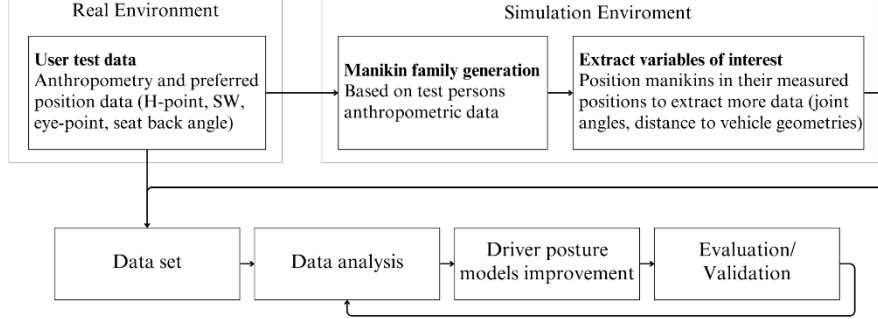


Figure 1. Schematic of the process.

2.1 Evaluation study

A test and evaluation study was conducted to demonstrate the practical application of the proposed method. The data for this study was sourced from an internal experimental study conducted by Volvo Cars in 2017, which involved 49 drivers. This study included an anthropometrically diverse group of 25 men and 24 women. The stature distribution within the test group was intentionally not normally distributed, where 16 participants (all women) were shorter than 1600 mm and 16 participants (all men) were taller than 1937mm (Figure 2). The participants were asked to select their preferred driving posture while driving an SUV model under real road conditions. Data collected during the study included participants' anthropometric measurements and driving position data. The driving position data included several key aspects, such as the seat position (H-point), seat back angle, steering wheel position, and eye point.

The simulation was modeled using the DHM tool IPS IMMA version 2023.R2 [7]. Utilizing the existing functionality in IPS IMMA, a manikin family was generated based on 13 anthropometric measurements from each test participant. Each manikin was then positioned according to the driving posture of the corresponding test participant, and all body joint angles and distance from the manikins to the roof, steering wheel, and instrument panel were recorded.

Following the schematic of the proposed process, the extracted additional variables, such as body joint angles, were analyzed to develop stature-based strategies. This involved determining the optimal joint angle values for driver posture predictions for each of the three groups: short, medium, and tall. Based on these optimal joint angle values per stature group, new strategies were created and evaluated against the "Seated Car" strategy defined in the IPS IMMA software [8].

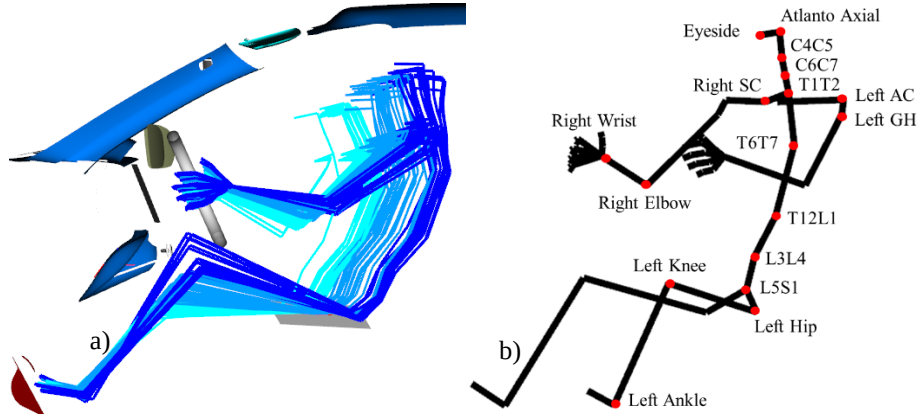


Figure 2. a) Manikin families: short (turquoise), medium (blue), and tall (dark blue); b) IPS IMMA skeleton and joint angle names.

3 Results

After extracting the joint angles, the data was categorized into three groups based on stature to examine variations in driving posture. The groups were defined as follows: short (1460 - 1622 mm), medium (1623 – 1936 mm), and tall (1937 - 2030 mm). Each group contained approximately 1/3 of the data and consisted of 16, 17, and 16 manikins, respectively. The analysis revealed significant differences in preferred joint angles across the three stature groups, with the most pronounced differences observed between the short and tall groups at the C6C7, hip, knee, and wrist joints. When compared to the "Seated Car" strategy of IPS IMMA, the set of predefined values for optimizing the driver posture in the software, the greatest discrepancies were found in the T12L1, hip, and knee joint angles. While differences in the ankle, elbow, and other torso joint angles were noted, they were less substantial compared to the aforementioned joints. These differences are illustrated in Figure 3.

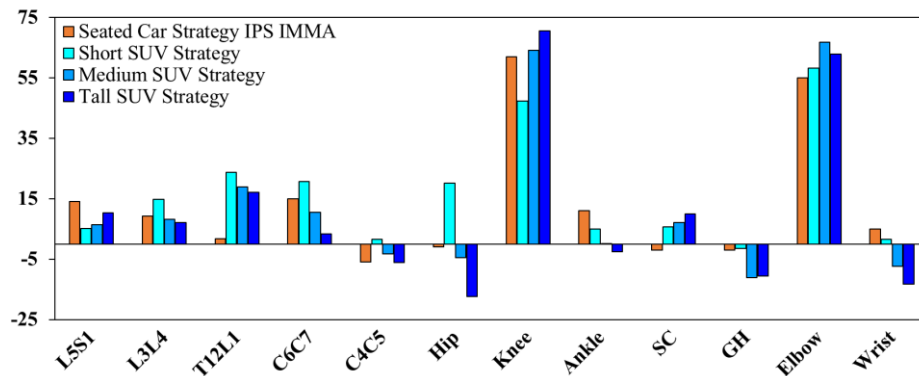


Figure 3. Comparison of simulation strategies angle values: Seated Car default (orange), Short (turquoise), Medium (blue), and Tall (dark blue).

3.1 Improvement of Driver Posture Models

The observed differences in joint angles among the stature groups were used to enhance driver posture prediction capabilities in the software. Three strategies were created based on the average joint angles of the manikins within each stature group. This resulted in a tailored set of preferred values for short, medium, and tall manikins. Then, simulations were done following the default “Seated Car” strategy in IPS IMMA and the developed Stature-based strategies based on the average of the extracted angles. Results from the simulation comparison between the strategies show smaller RMSE values on the stature-based strategies compared to the “Seated Car” in the eye- and h-point in the z-axis for the short and tall stature groups. However, the RMSE values, except for the eye-point x, are lower with the “Seated Car” strategy for the medium stature group, indicating that the software provides good predictions for this specific group. These results are described in Figure 4 and Table 1.

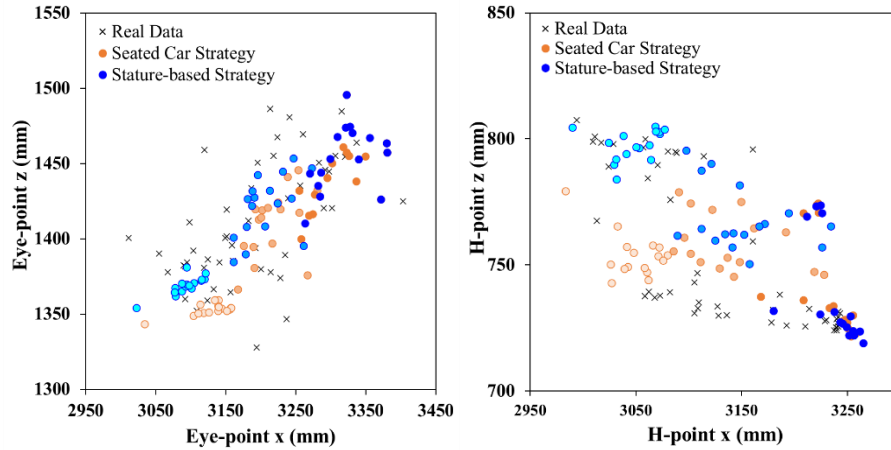


Figure 4. Predicted eye- and h-point positions by strategy and stature group: Seated Car (light orange, mid orange, orange) and Stature-based (turquoise, blue, dark blue) correspond to short, medium, and tall manikins respectively.

Table 1. RMSE (mm) by strategy and stature group.

	H-point x		H-point z		Eye-point x		Eye-point z	
	Seated Car	Stature-based	Seated Car	Stature-based	Seated Car	Stature-based	Seated Car	Stature-based
Short	23.6	21.9	36.4	22.9	55.9	52.2	32.3	22.8
Medium	40.7	44.2	23.8	29.9	44.2	41.5	32.0	36.2
Tall	38.4	42.7	23.4	23.1	93.7	94.6	27.1	22.9

4 Discussion

The proposed integrated approach in this study has the potential to improve the prediction and evaluation of vehicle ergonomics. Leveraging data from previous driver posture studies maximizes the utility of available resources. Data from a constrained

environment can be considered valuable for developing strategies even in a less constrained setting due to the original simulation's limited diversity. This approach improves understanding of human-system interaction and predictive accuracy by continuously integrating new data from driver posture studies into the model.

Despite the potential benefits, several limitations exist that need further investigation. While the proposed process enhances predictive accuracy, there is room for improvement in refining manikin body meshes and considering seat properties within the DHM software. Incorporating biomechanical models to calculate seat deflection could further enhance the accuracy of simulations. Additionally, other parameters of how the optimization model runs in the software could be refined to obtain more realistic results. It should also be noted that in this initial study, only data from one SUV vehicle model was analyzed and used to develop the strategies. Validation with more vehicle models would be necessary to ensure the robustness and generalizability of the driver posture models. Addressing these limitations would optimize the outcomes of the proposed process and provide more comprehensive results.

In conclusion, the proposed process presents a promising solution for incomplete datasets in occupant packaging studies and provides researchers with a valuable tool for enhancing analysis and creating specific driver posture prediction models.

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